

Brennan Duff
Assignment 3

1. Retrieve the plant name, country name, operator name, fuel category, fuel name, capacity (in MW), and commission year for all power plants without using table aliases. Sort the results in descending order by capacity.

a. `SELECT power_plants.plantname,`
`countries.countryname,`
`operators.operatorname,`
`fuel_types.fuelcategory,`
`fuel_types.fuelname,`
`power_plants.capacitymw,`
`power_plants.commissionyear`
`FROM power_plants`
`JOIN countries`
`ON power_plants.countrycode = countries.countrycode`
`JOIN operators`
`ON power_plants.operatorid = operators.operatorid`
`JOIN fuel_types`
`ON power_plants.fuelid = fuel_types.fuelid`
`ORDER BY power_plants.capacitymw DESC;`

b.

	plantname	countryname	operatorname	fuelcategory	fuelname	capacitymw	commissionyear
►	Three Gorges	China	China Yangtze Power	Renewable	Hydro	22500	2003
	Itaipu	Brazil	Electrobras	Renewable	Hydro	14000	1984
	Tuoketuo	China	China Yangtze Power	Fossil	Coal	13200	1995
	Kashiwazaki-Kariwa	Japan	Tepco Energy	Nuclear	Uranium	8212	1985
	Grand Coulee	United States	Global Hydro Inc	Renewable	Hydro	6809	1942
	Vindhyachal	India	NTPC Limited	Fossil	Coal	4760	1987
	Janschwalde	Germany	Rhein Power AG	Fossil	Coal	3000	1981
	Loy Yang	Australia	Snowy Hydro	Fossil	Coal	2210	1984

2. Retrieve the plant name, country code, calendar year, and annual generation (in GWh) for all power plants for the year 2024 without using table aliases. Sort the results in descending order by generation.

```
a. SELECT power_plants.plantname,
        power_plants.countrycode,
        generation_records.year,
        generation_records.generationgwh
FROM power_plants
JOIN generation_records
    ON power_plants.plantid = generation_records.plantid
WHERE generation_records.year = 2024
ORDER BY generation_records.generationgwh DESC;
```

	plantname	countrycode	year	generationgwh
►	Three Gorges	CHN	2024	98800
	Itaipu	BRA	2024	89000.1
	Tuocketuo	CHN	2024	71000.3
	Vindhyachal	IND	2024	35600.8
	Grand Coulee	USA	2024	21500.5
	Janschwalde	DEU	2024	18400.2
	Loy Yang	AUS	2024	14200.4
b.	Kashiwazaki-Kariwa	JPN	2024	0

3. Retrieve the plant name, country code, calendar year, annual generation (in GWh), and CO2 emissions (in tonnes) for all power plants for the year 2024 without using table aliases. Sort the results in ascending order by CO2 emissions.

```
a. SELECT power_plants.plantname,
        power_plants.countrycode,
        generation_records.year,
```

```

generation_records.generationgwh,
emission_metrics.co2emissionstonnes

FROM power_plants

JOIN generation_records

ON power_plants.plantid = generation_records.plantid

JOIN emission_metrics

ON generation_records.plantid = emission_metrics.plantid

AND generation_records.year = emission_metrics.year

WHERE generation_records.year = 2024

ORDER BY emission_metrics.co2emissionstonnes ASC;

```

	plantname	countrycode	year	generationgwh	co2emissionstonnes
►	Grand Coulee	USA	2024	21500.5	0
	Itaipu	BRA	2024	89000.1	0
	Kashiwazaki-Kariwa	JPN	2024	0	0
	Three Gorges	CHN	2024	98800	0
	Loy Yang	AUS	2024	14200.4	1950000
	Janschwalde	DEU	2024	18400.2	2450000
	Vindhyachal	IND	2024	35600.8	3100000
b.	Tuocketuo	CHN	2024	71000.3	5200000

4. Write a SQL query using a **Common Table Expression (CTE)** to calculate the total cumulative power generation (in GWh) for each operator across all available years without using table aliases. Retrieve the operator name, headquarters country, and their total generated power, and sort the results in descending order by total generation.

```

a. WITH total_generation AS (

SELECT power_plants.operatorid,

SUM(generation_records.generationgwh) AS totalgeneration

FROM power_plants

```

```

JOIN generation_records

    ON power_plants.plantid = generation_records.plantid

GROUP BY power_plants.operatorid
)

SELECT operators.operatorname,

    operators.headquarterscountry,

    total_generation.totalgeneration

FROM total_generation

JOIN operators

    ON total_generation.operatorid = operators.operatorid

ORDER BY total_generation.totalgeneration DESC;

```

	operatorname	headquarterscountry	totalgeneration
►	China Yangtze Power	CHN	169800.3
	Electrobras	BRA	89000.1
	NTPC Limited	IND	35600.8
	Global Hydro Inc	USA	21500.5
	Rhein Power AG	DEU	18400.2
	Snowy Hydro	AUS	14200.4
b.	Tepco Energy	JPN	0

5. Write a SQL query using **two separate Common Table Expressions (CTEs)**--one to calculate total power generation by country code and another to calculate total CO2 emissions by country code without using table aliases. Then, join these CTEs with the countries table to display the country name, total generation (in GWh), and total CO2 emissions (in tonnes). Sort the results in descending order by total generation.

- a. WITH country_generation AS (
- ```

SELECT power_plants.countrycode,

```

```

 SUM(generation_records.generationgwh) AS totalgeneration
FROM power_plants
JOIN generation_records
 ON power_plants.plantid = generation_records.plantid
GROUP BY power_plants.countrycode
),
country_emissions AS (
 SELECT power_plants.countrycode,
 SUM(emission_metrics.co2emissionstonnes) AS totalemissions
 FROM power_plants
 JOIN emission_metrics
 ON power_plants.plantid = emission_metrics.plantid
 GROUP BY power_plants.countrycode
)
SELECT countries.countryname,
 country_generation.totalgeneration,
 country_emissions.totalemissions
FROM countries
JOIN country_generation
 ON countries.countrycode = country_generation.countrycode
JOIN country_emissions
 ON countries.countrycode = country_emissions.countrycode
ORDER BY country_generation.totalgeneration DESC;

```

|   | countryname   | totalgeneration | totalemissions |
|---|---------------|-----------------|----------------|
| ▶ | China         | 169800.3        | 5200000        |
|   | Brazil        | 89000.1         | 0              |
|   | India         | 35600.8         | 3100000        |
|   | United States | 21500.5         | 0              |
|   | Germany       | 18400.2         | 2450000        |
|   | Australia     | 14200.4         | 1950000        |
|   | Japan         | 0               | 0              |

b.

|   |    |          |                                                                                                  |                   |                       |
|---|----|----------|--------------------------------------------------------------------------------------------------|-------------------|-----------------------|
| ✓ | 50 | 20:17:45 | SELECT power_plants.plantname, countries.countryname, operators.operatorname, fuel_types.fu...   | 8 row(s) returned | 0.016 sec / 0.000 sec |
| ✓ | 51 | 20:17:46 | SELECT power_plants.plantname, power_plants.countrycode, generation_records.year, generati...    | 8 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ | 52 | 20:17:46 | SELECT power_plants.plantname, power_plants.countrycode, generation_records.year, generati...    | 8 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ | 53 | 20:17:46 | WITH total_generation AS ( SELECT power_plants.operatorid, SUM(generation_records.generationg... | 7 row(s) returned | 0.000 sec / 0.000 sec |
| ✓ | 54 | 20:17:46 | WITH country_generation AS ( SELECT power_plants.countrycode, SUM(generation_records.gener...    | 7 row(s) returned | 0.000 sec / 0.000 sec |

